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Digital assistant interactions in extended reality

Abstract

An example process includes while displaying, on a display, an extended reality (XR) environment: receiving a user input; sampling, with a microphone, a user speech input; in accordance with a determination that the user input satisfies a criterion for initiating a digital assistant, initiating the digital assistant, including: displaying, within the XR environment, a digital assistant indicator at a first location of the XR environment; and while displaying the digital assistant indicator at the first location, providing, by the digital assistant, a response to the user speech input; after providing the response, ceasing to display the digital assistant indicator at the first location; and in accordance with ceasing to display the digital assistant indicator at the first location, displaying the digital assistant indicator at a second location of the XR environment, the second location corresponding to a physical location of an external electronic device implementing a second digital assistant.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a U.S. National Stage Patent Application of PCT Application No. PCT/US2022/017343, entitled “DIGITAL ASSISTANT INTERACTIONS IN EXTENDED REALITY,” filed on Feb. 22, 2022, which claims priority to U.S. Patent Application No. 63/152,723, entitled “DIGITAL ASSISTANT INTERACTIONS IN EXTENDED REALITY,” filed on Feb. 23, 2021. The contents of each of these applications are hereby incorporated by reference in their entireties.

FIELD

(1) This relates generally to using digital assistants with extended reality technologies.

BACKGROUND

(2) Digital assistants may allow users to interact with devices using natural language input in spoken or text form. For example, a user can provide a speech input including a user request to a digital assistant operating on an electronic device. The digital assistant can determine the user's intent from the speech input and determine a corresponding task. The digital assistant can perform the task by executing one or more services of the electronic device to provide an output responsive to the user request.

SUMMARY

(3) Example methods are disclosed herein. An example method includes: while displaying, on a display of an electronic device, an extended reality (XR) environment: receiving, from a user, a user input; sampling, with a microphone of the electronic device, a user speech input; in accordance with a determination that the user input satisfies a criterion for initiating a digital assistant, initiating the digital assistant, including: displaying, within the XR environment, a digital assistant indicator at a first location of the XR environment; and while displaying the digital assistant indicator at the first location, providing, by the digital assistant, a response to the user speech input; after providing the response, ceasing to display the digital assistant indicator at the first location; and in response to ceasing to display the digital assistant indicator at the first location, displaying the digital assistant indicator at a second location of the XR environment, the

second location corresponding to a physical location of an external electronic device implementing a second digital assistant.

(4) Example non-transitory computer-readable media are disclosed herein. An example non-transitory computer-readable storage medium stores one or more programs. The one or more programs comprise instructions, which when executed by one or more processors of an electronic device having a display and a microphone, cause the electronic device to: while displaying, on the display, an extended reality (XR) environment: receive, from a user, a user input; sample, with the microphone, a user speech input; in accordance with a determination that the user input satisfies a criterion for initiating a digital assistant, initiate the digital assistant, including: displaying, within the XR environment, a digital assistant indicator at a first location of the XR environment; and while displaying the digital assistant indicator at the first location, providing, by the digital assistant, a response to the user speech input; after providing the response, cease to display the digital assistant indicator at the first location; and in response to ceasing to display the digital assistant indicator at the first location, display the digital assistant indicator at a second location of the XR environment, the second location corresponding to a physical location of an external electronic device implementing a second digital assistant.

(5) Example electronic devices are disclosed herein. An example electronic device comprises a display, a microphone, one or more processors; a memory; and one or more programs, where the one or more programs are stored in the memory and configured to be executed by the one or more processors, the one or more programs including instructions for: while displaying, on the display, an extended reality (XR) environment: receiving, from a user, a user input; sampling, with the microphone, a user speech input; in accordance with a determination that the user input satisfies a criterion for initiating a digital assistant, initiating the digital assistant, including: displaying, within the XR environment, a digital assistant indicator at a first location of the XR environment; and while displaying the digital assistant indicator at the first location, providing, by the digital assistant, a response to the user speech input; after providing the response, ceasing to display the digital assistant indicator at the first location; and in response to ceasing to display the digital assistant indicator at the first location, displaying the digital assistant indicator at a second location of the XR environment, the second location corresponding to a physical location of an external electronic device implementing a second digital assistant.

(6) An example electronic device comprises, means for: while displaying, on a display of the electronic device, an extended reality (XR) environment: receiving, from a user, a user input; sampling, with the microphone of the electronic device, a user speech input; in accordance with a determination that the user input satisfies a criterion for initiating a digital assistant, initiating the digital assistant, including: displaying, within the XR environment, a digital assistant indicator at a first location of the XR environment; and while displaying the digital assistant indicator at the first location, providing, by the digital assistant, a response to the user speech input; after providing the response, ceasing to display the digital assistant indicator at the first location; and in response to ceasing to display the digital assistant indicator at the first location, displaying the digital assistant indicator at a second location of the XR environment, the second location corresponding to a physical location of an external electronic device implementing a second digital assistant.

(7) Displaying the digital assistant indicator at the second location can provide, to a user immersed in an XR experience, the impression of a digital assistant being present at (or near) a physical location known to the user (e.g., the location of the external electronic device). Indicating the presence of a digital assistant at a known physical location can improve the user's spatial awareness of their physical environment and enable intuitive and efficient interactions with the digital assistant. For example, to initiate an interaction with the digital assistant, the user can issue a spoken request to the digital assistant while facing the known physical location. The user is not required to provide other inputs to initiate the interaction (e.g., inputs explicitly indicating to initiate a digital assistant such as a spoken trigger). In this manner, the user-device interface is

made more efficient and accurate (e.g., by reducing the time and inputs required to initiate digital assistant interactions, by accurately initiating digital assistant interactions when desired, by reducing user inputs to disable incorrectly initiated digital assistant interactions), which, additionally, reduces power usage and improves battery life of the device.

Description

BRIEF DESCRIPTION OF FIGURES

- (1) FIGS. 1A-1B depict exemplary systems for use in various extended reality technologies.
- (2) FIG. 2 illustrates a block diagram of a system for providing digital assistant interactions, according to various examples.
- (3) FIGS. 3A-3B illustrate digital assistant interactions within an XR environment, according to various examples.
- (4) FIG. 4 is a flow diagram illustrating a process for digital assistant interaction, according to various examples.

DESCRIPTION

(5) People may sense or interact with a physical environment or world without using an electronic device. Physical features, such as a physical object or surface, may be included within a physical environment. For instance, a physical environment may correspond to a physical city having physical buildings, roads, and vehicles. People may directly sense or interact with a physical environment through various means, such as smell, sight, taste, hearing, and touch. This can be in contrast to an extended reality (XR) environment that may refer to a partially or wholly simulated environment that people may sense or interact with using an electronic device. The XR environment may include virtual reality (VR) content, mixed reality (MR) content, augmented reality (AR) content, or the like. Using an XR system, a portion of a person's physical motions, or representations thereof, may be tracked and, in response, properties of virtual objects in the XR environment may be changed in a way that complies with at least one law of nature. For example, the XR system may detect a user's head movement and adjust auditory and graphical content presented to the user in a way that simulates how sounds and views would change in a physical environment. In other examples, the XR system may detect movement of an electronic device (e.g., a laptop, tablet, mobile phone, or the like) presenting the XR environment. Accordingly, the XR system may adjust auditory and graphical content presented to the user in a way that simulates how sounds and views would change in a physical environment. In some instances, other inputs, such as a representation of physical motion (e.g., a voice command), may cause the XR system to adjust properties of graphical content.

(6) Numerous types of electronic systems may allow a user to sense or interact with an XR environment. A non-exhaustive list of examples includes lenses having integrated display capability to be placed on a user's eyes (e.g., contact lenses), heads-up displays (HUDs), projection-based systems, head mountable systems, windows or windshields having integrated display technology, headphones/earphones, input systems with or without haptic feedback (e.g., handheld or wearable controllers), smartphones, tablets, desktop/laptop computers, and speaker arrays. Head mountable systems may include an opaque display and one or more speakers. Other head mountable systems may be configured to receive an opaque external display, such as that of a smartphone. Head mountable systems may capture images/video of the physical environment using one or more image sensors or capture audio of the physical environment using one or more microphones. Instead of an opaque display, some head mountable systems may include a transparent or translucent display. Transparent or translucent displays may direct light representative of images to a user's eyes through a medium, such as a hologram medium, optical waveguide, an optical combiner, optical reflector, other similar technologies, or combinations

thereof. Various display technologies, such as liquid crystal on silicon, LEDs, uLEDs, OLEDs, laser scanning light source, digital light projection, or combinations thereof, may be used. In some examples, the transparent or translucent display may be selectively controlled to become opaque. Projection-based systems may utilize retinal projection technology that projects images onto a user's retina or may project virtual content into the physical environment, such as onto a physical surface or as a hologram.

(7) FIGS. 1A-1B depict exemplary system **100** for use in various extended reality technologies.

(8) As shown in FIG. 1A, system **100** includes device **100a**. Device **100a** includes RF circuitry(ies) **104**, processor(s) **102**, memory(ies) **106**, image sensor(s) **108**, touch-sensitive surface(s) **122**, speaker(s) **118**, location sensor(s) **116**, microphone(s) **112**, orientation sensor(s) **110**, and display(s) **120**. These components optionally communicate using communication bus(es) **150** of device **100a**.

(9) In some examples, a base station device (e.g., a computing device, such as a remote server, mobile device, or laptop) implements some components of system **100** and a second device (e.g., a head-mounted device) implements other components of system **100**. In some examples, device **100a** is implemented in a base station device or in a second device.

(10) As shown in FIG. 1B, in some examples, system **100** includes two or more devices in communication, e.g., via a wired connection or a wireless connection. First device **100b** (e.g., a base station device) includes memory(ies) **106**, RF circuitry(ies) **104**, and processor(s) **102**. Such components optionally communicate using communication bus(es) **150** of device **100b**. Second device **100c** (e.g., a head-mounted device) includes components such as RF circuitry(ies) **104**, processor(s) **102**, memory(ies) **106**, image sensor(s) **108**, touch-sensitive surface(s) **122**, speaker(s) **118**, location sensor(s) **116**, microphone(s) **112**, orientation sensor(s) **110**, and display(s) **120**. These components optionally communicate using communication bus(es) **150** of device **100c**.

(11) System **100** includes RF circuitry(ies) **104**. RF circuitry(ies) **104** optionally include circuitry for communicating with networks (e.g., the Internet, a wireless network (e.g., such as cellular networks and wireless local area networks (LANs)), and/or intranets) and/or electronic devices. RF circuitry(ies) **104** optionally includes circuitry for communicating using near-field communication and/or short-range communication (e.g., Bluetooth®).

(12) System **100** includes processor(s) **102** and memory(ies) **106**. Processor(s) **102** include one or more graphics processors, one or more general processors, and/or one or more digital signal processors. In some examples, memory(ies) **106** are one or more non-transitory computer-readable storage mediums (e.g., random access memory, flash memory) storing computer-readable instructions configured to be executed by processor(s) **102** to perform the techniques described below.

(13) System **100** includes image sensor(s) **108**. Image sensors(s) **108** optionally include one or more infrared (IR) sensor(s), e.g., a passive IR sensor or an active IR sensor, to detect infrared light from the physical environment. For example, an active IR sensor includes an IR emitter (e.g., an IR dot emitter) for emitting infrared light into the physical environment. Image sensor(s) **108** also optionally include one or more visible light image sensors, such as complementary metal-oxide-semiconductor (CMOS) sensors and/or charged coupled device (CCD) sensors capable of obtaining images of physical elements from the physical environment. Image sensor(s) **108** also optionally include one or more event camera(s) configured to capture movement of physical elements in the physical environment. Image sensor(s) **108** also optionally include one or more depth sensor(s) capable of detecting the distance of physical elements from system **100**. In some examples, system **100** uses IR sensors, CCD sensors, event cameras, and depth sensors together to detect the physical environment around system **100**. In some examples, image sensor(s) **108** include first and second image sensors. The first and second image sensors are optionally capable of capturing images of physical elements in the physical environment from two respective different perspectives. In some examples, system **100** uses image sensor(s) **108** to detect the position and orientation of system **100** and/or display(s) **120** in the physical environment. For example, system **100** uses image sensor(s)

108 to track the position and orientation of display(s) **120** relative to one or more fixed elements in the physical environment. In some examples, image sensor(s) **108** are capable of receiving user inputs, such as hand gestures.

(14) In some examples, system **100** includes touch-sensitive surface(s) **122** for receiving user inputs, such as tapping or swiping inputs. In some examples, touch-sensitive surface(s) **122** and display(s) **120** are combined into touch-sensitive display(s).

(15) In some examples, system **100** includes microphones(s) **112**. System **100** uses microphone(s) **112** to detect sound from the user's physical environment or from the user. In some examples, microphone(s) **112** includes a microphone array (e.g., including a plurality of microphones) that optionally operate together, e.g., to locate the spatial source of sound from the physical environment or to identify ambient noise.

(16) System **100** includes orientation sensor(s) **110** for detecting orientation and/or movement of system **100** and/or display(s) **120**. For example, system **100** uses orientation sensor(s) **110** to track changes in the position and/or orientation of system **100** and/or display(s) **120**, such as relative to physical elements in the physical environment. Orientation sensor(s) **110** optionally include gyroscope(s) and/or accelerometer(s).

(17) System **100** includes display(s) **120**. Display(s) **120** may operate with a transparent or semi-transparent displays (and optionally with one or more imaging sensors). Display(s) **120** may include an opaque display. Display(s) **120** may allow a person to view a physical environment directly through the display, and may also allow addition of virtual content to the person's field of view, e.g., by superimposing virtual content over the physical environment. Display(s) **120** may implement display technologies such as a digital light projector, a laser scanning light source, LEDs, OLEDs, liquid crystal on silicon, or combinations thereof. Display(s) **120** can include substrates through which light is transmitted, e.g., optical reflectors and combiners, light waveguides, holographic substrates, or combinations thereof. As a particular example, the transparent or semi-transparent display may selectively transition between a transparent or semi-transparent state and an opaque state. Further example implementations of display(s) **120** include display-capable lenses, tablets, smartphones, desktop computers, laptop computers, heads up displays, display-capable automotive windshields, or display-capable windows. In some examples, system **100** is a projection-based system. For example, system **100** projects virtual objects onto a physical environment (e.g., projects a holograph onto a physical environment or projects imagery onto a physical surface). As another example, system **100** uses retinal projection to project images onto a person's eyes (e.g., retina). In some examples, system **100** can be configured to interface with an external display (e.g., a smartphone display).

(18) FIG. 2 illustrates a block diagram of system **200** for providing digital assistant (DA) interactions, according to various examples.

(19) System **200** includes DA **202**. In some examples, as shown, DA **202** is at least partially implemented within system **100**, e.g., within device **100a**, **100b**, or **100c**. For example, DA **202** is at least partially implemented as computer-executable instructions stored in memory(ies) **106**. In some examples, DA **202** is distributed across multiple computers and/or systems. In some examples, some of the modules and functions of DA **202** are divided into a server portion and a client portion, where the client portion is implemented in one or more user devices (e.g., device **100a**, **100b**, **100c**) or systems (e.g., system **100**) and, optionally, communicates with the server portion via one or more networks. The various components and functions of DA **202** are implemented in hardware, software instructions for execution by one or more processors, firmware, including one or more signal processing and/or application specific integrated circuits, or a combination thereof. DA **202** is only one example of a DA, and DA **202** can have more or fewer components than shown, can combine two or more components, or can have a different configuration or arrangement of the components.

(20) In some examples, DA **202** performs at least some of: converting speech input into text,

identifying a user's intent from a received natural language input, eliciting (e.g., from the user) and obtaining information needed to fully satisfy the user's intent (e.g., by disambiguating phrases, names, etc.), determining a task flow for satisfying the identified intent, and executing the task flow.

(21) In some examples, DA **202** includes natural language processor **206** configured to identify the user intent. Natural language processor **206** takes the candidate textual representations generated by speech-to-text (STT) processor **204**, and attempts to map each of the candidate textual representations to one or more “intents” recognized by the DA. An “intent” (or “user intent”) represents a DA performable task, and can have a corresponding task flow implemented in task flow processor **208**. The corresponding task flow is a series of programmed actions that the DA takes for performing the task. Accordingly, in some examples, the DA's capabilities are dependent on the types of different task flows implemented in task flow processor **208**, e.g., dependent on the different types of “intents” the DA recognizes.

(22) In some examples, once natural language processor **206** identifies an intent based on the natural language input, natural language processor **206** causes task flow processor **208** to perform the actions required to satisfy the user request. For example, task flow processor **208** executes the task flow corresponding to the identified intent to perform a task to satisfy the user request. In some examples, performing the task includes providing audible, haptic, or displayed output indicating the results of the performed task.

(23) System **200** includes device **210**. Device **210** is external to a device displaying an XR environment. The architecture of external device **210** is similar or identical to that of device **100b** or **100c**. For example, external device **210** includes processor(s), memory(ies), RF circuitry(ies), communication bus(es), speaker(s), microphone(s), display(s), or a combination or sub-combination thereof. In some examples, external device **210** is implemented as a smart speaker. In other examples, external device **210** is implemented as a smartphone, laptop or desktop computer, tablet device, wearable device (e.g., smart watch), a smart home device (smart TV, smart appliance), or combination or sub-combination thereof. In some examples, external device **210** communicates with other devices (e.g., system **100**) through a wired or wireless connection.

(24) External device **210** implements, at least partially, DA **212**. The architecture of DA **212** is similar or identical to that of DA **202**. For example, while FIG. 2 shows that DA **212** is implemented on a single device (e.g., external device **210**), in other examples, DA **212** is distributed across multiple computers and/or systems, e.g., as a server portion and a client portion. DA **212** is capable of performing all of the functions described above with respect to DA **202**, e.g., STT conversion, natural language processing, and task flow execution.

(25) FIGS. 3A-3B illustrate DA interactions within an XR environment, according to various examples.

(26) FIG. 3A shows user **302** immersed in an XR experience using device **304**, e.g., a head mounted device worn by user **302**. Device **304** is implemented as device **100a** or **100c**. The right panels of FIGS. 3A-3B shows display **306** of device **304**, e.g., the current view of user **302**.

(27) In FIG. 3A, user **302** is located in physical environment **300**, e.g., a physical room including a physical painting, a physical table, and external device **210**. Although user **302** faces the front of physical environment **300**, user **302** views XR environment **308** displayed on display **306**.

(28) User **302** sometimes desires to interact with a DA while immersed in the XR experience. Accordingly, user **302** provides a user input to initiate the DA. For example, in FIG. 3A, user **302** provides the spoken trigger “Hey Assistant.”

(29) In some examples, while displaying XR environment **308**, device **304** receives the user input and determines whether the user input satisfies a criterion for initiating a DA. Example user input satisfying the criterion for initiating a DA include explicit triggers to initiate the DA, such as button input (e.g., input selecting a button of device **304**, or input selecting the button for a predetermined duration), speech input (e.g., speech including a spoken trigger for initiating a DA, e.g., “Hey

Assistant,” “Wake up,” and the like), gesture input (e.g., predetermined types of gestures, e.g., hand gestures, arm motions, tapping motions, and the like), motion input (e.g., motion of device **304** representing predetermined user movements, e.g., head nods, head shakes, and the like), and input received at an external device communicatively coupled to device **304** (e.g., a press of a button of the external device, input moving the external device in a predetermined manner).

(30) In FIG. **3A**, user **302** further provides a speech input including a request to the DA. For example, user **302** asks “is it cold out?” after providing the user input to initiate the DA. Device **304** samples the speech input.

(31) In some examples, in accordance with a determination that the user input satisfies a criterion for initiating the DA, device **304** initiates the DA, e.g., DA **202**. For example, device **304** displays DA indicator **310** within XR environment **308** to indicate the initiated DA. In some examples, device **304** displays DA indicator **310** at a first location of the XR environment that is visible from user **302**'s current pose (e.g., head pose). For example, whenever user **302** initiates a DA using an explicit trigger, device **304** displays DA indicator **310** at a same location on display **306**, e.g., using the same plurality of pixels.

(32) In some examples, initiating the DA further includes providing, by the DA, a response to the user speech input. For example, while displaying DA indicator **310** at the first location, device **304** speaks the response “70 degrees today, so it's not cold outside” to satisfy the user request.

(33) Requiring user **302** to provide an explicit trigger to initiate each DA interaction may be burdensome, non-intuitive, and interrupt the user's XR experience. The below describes techniques for using external device **210** to initiate DA interactions without requiring user **302** to provide explicit triggers to initiate the interactions.

(34) In some examples, after providing the response to the user speech input, device **304** ceases to display DA indicator **310** at the first location, e.g., to indicate that the current DA interaction has ended. In some examples, device **304** ceases to display DA indicator **310** automatically (e.g., without user input instructing to cease the display) at a predetermined duration after providing the response. For example, device **304** automatically ceases to display DA indicator **310** at the end of the predetermined duration in accordance with not receiving a further user request to the DA (e.g., not sampling follow-up speech input and/or not receiving another explicit trigger) before the predetermined duration elapses. In some examples, device **304** ceases to display DA indicator **310** responsive to receiving a user input representing an instruction to dismiss the display of DA indicator **310**, e.g., user input to end the current DA interaction. In some examples, the user input includes gesture input (e.g., a predetermined gesture performed by user **302** to end the current DA interaction) or speech input (e.g., speech inputs such as “I'm done” or “go away now” to end the current DA interaction). For example, while device **304** displays DA indicator **310** at the first location, device **304** samples the speech input “go away,” and in response, ceases to display DA indicator **310** at the first location.

(35) In some examples, in accordance with ceasing to display DA indicator **310** at the first location, device **304** displays DA indicator **310** at a second location of XR environment **308**. The second location corresponds to the physical location of external device **210**, e.g., the location of external device **210** in XR environment **308** if external device **210** were displayed via pass-through video. For example, the second location is within a predetermined distance of the physical location, e.g., within a sphere defined by the predetermined distance from the physical location. As one example, the second location is directly above the physical location, e.g., is above the physical location on the vertical axis of external device **210**, as shown in FIG. **3B**. In this manner, device **304** may move the display of DA indicator **310** to a location known by user **302**, as user **302** knows where external device **210** (e.g., the user's smart speaker) is located.

(36) In some examples, the gesture input representing the instruction to dismiss the display of DA indicator **310** at the first location includes a drag-and-drop gesture from the first location to the second location. For example, device **304** detects a gesture input from user **302** including a press

gesture (e.g., a finger press gesture) corresponding to the first location, a dragging gesture (e.g., a finger drag gesture) corresponding to movement of DA indicator **310** from the first location to the second location, and a release gesture (e.g., a finger release gesture) corresponding to the second location. In this manner, user **302** can select DA indicator **310** at the first location and move DA indicator **310** to the second location corresponding to the location of a user-selected external electronic device. Accordingly, if user **302** owns multiple external electronic devices each implementing respective DAs (e.g., a smart phone, a smart watch, multiple smart speakers), user **302** can select a particular external electronic device (e.g., external device **210**) for the display of DA indicator **310**.

(37) If multiple external electronic devices (e.g., each implementing DAs) are available for the display of DA indicator **310**, device **304** may consider various other criteria to select the external device corresponding to the second location, e.g., if user gesture input does not indicate the external device. For example, device **304** ranks user **302**'s external devices according to various criteria and selects the top-ranked device. Example criteria include external device type, whether the external device is a user-designated default device, frequency and/or recency of user interaction with a DA implemented on the external device, and location of the external device relative to device **304**. For example, device **304** ranks certain external device types (e.g., smart speakers, smart home appliances) higher than other external device types (e.g., smart phones and smart watches). As another example, device **304** ranks a user-designated default external device higher than other external devices. As yet another example, device **304** ranks an external device higher if user **302** most frequently and/or most recently interacted with a DA implemented on the external device. As yet another example, device **304** ranks external devices that are physically closer to device **304** higher than external devices physically further from device **304**. In some examples, device **304** considers some or all of the above-described criteria when ranking external devices, e.g., by assigning each of the above-described criteria a respective weight when ranking the external devices.

(38) Displaying DA indicator **310** at the known (second) location, or in other words, anchoring the display of DA indicator **310** to the physical location of external device **210**, can orient user **302** to physical environment **300** while the user is immersed in an XR experience. For example, viewing DA indicator **310** (e.g., in FIG. 3B) allows user **302** to determine their current physical position and orientation relative to external device **210**. Displaying DA indicator **310** at the known location further indicates the availability of a DA at the known location, which may provide an efficient and intuitive manner for initiating interactions with the DA. For example, as discussed below, user **302** may simply face the DA (e.g., face the known location) and provide a spoken request to initiate a DA interaction, e.g., without providing an explicit trigger to initiate the interaction.

(39) In some examples, device **304** displays DA indicator **310** at the second location further in accordance with determining that the second location is visible from user **302**'s current pose. For example, device **304** determines whether external device **210** is visible from the current pose (and thus whether the second location is visible from the current pose). In some examples, device **304** determines that external device **210** is visible from the current pose by detecting (e.g., locating) external device **210** using one or more sensors of device **304** when user **302** has the current pose. For example, device **304** detects external device **210** within a region representing (or approximating) user **302**'s field of view in the current pose, e.g., user **302**'s field of view of physical environment **300** if user **302** were not wearing device **304**.

(40) In some examples, the sensor(s) of device **304** include a camera. In some examples, device **304** analyzes image data captured by the camera to recognize external device **210** (e.g., using object recognition technologies) and locate external device **210** relative to device **304**. In some examples, the sensor(s) of device **304** includes sensor(s) configured to detect external device **210** via signals emitted by external device **210** and/or device **304** (e.g., sensors using range finding technology). In some examples, device **304** analyzes the signals to locate external device **210**

relative to device **304**. In this manner, device **304** can detect external device **210** within particular defined regions (e.g., within a first region representing user **302**'s current field of view, within a narrower front-facing region representing the portion of physical environment **300** that user **302** currently faces) to determine whether the second location is visible from user **302**'s current pose. (41) In some examples, device **304** further displays, within XR environment **308**, a physical representation of external device **210** at the physical location of external device **210**, e.g., via pass-through video. In some examples, device **304** similarly displays the physical representation of external device **210** in accordance with determining that external device **210** is visible from user **302**'s current pose.

(42) In the example of FIG. 3A, external device **210** is not visible from user **302**'s current pose. For example, user **302** faces the front of physical environment **300**, but external device **210** is located at the right side of physical environment **300**. FIG. 3B shows that user **302** has changed their pose to face (e.g., turned to face) the right side of physical environment **300**. For example, after the DA interaction of FIG. 3A ends (e.g., after device **304** ceases to display DA indicator **310** at the first location), user **302** turns to face external device **210** to initiate another DA interaction.

(43) In FIG. 3B, device **304** updates the displayed view of XR environment **308** to correspond to user **302**'s changed pose. Further, device **304** determines that external device **210** and the second location are visible from user **302**'s pose in FIG. 3B. Accordingly, device **304** displays external device **210** at its physical location and displays DA indicator **310** at the second location.

(44) In some examples, in accordance with determining that the second location is visible from user **302**'s current pose, device **304** displays (e.g., continues to display) virtual elements of XR environment **308** concurrently with DA indicator **310**. For example, FIG. 3B shows that device **304** modifies the view of XR environment **308** (relative to FIG. 3A) to correspond to user **302**'s right facing pose. The view includes a virtual tree (e.g., included in the right side of XR environment **308**) in addition to DA indicator **310**. In other examples, in accordance with determining that the second location is visible from user **302**'s current pose, device **304** displays DA indicator **310** as the only virtual element. For example, device **304** displays a view of physical environment **300** (e.g., displays the right side of the physical room via pass-through video), where the view includes DA indicator **310**, but does not include any other virtual element such as the virtual tree.

(45) In some examples, device **304** maintains the display of DA indicator **310** at the second location responsive to changes in user **302**'s pose. In this manner, device **304** keeps the display of DA indicator **310** anchored to the physical location of external device **210**. For example, device **304** updates the display of XR environment **308** based on updates to user **302**'s pose, but maintains the display of DA indicator **310** at the second location while updating the display of XR environment **308**. For example, if user **302** turns right in FIG. 3B, device **304** displays virtual elements to the right of the virtual tree, but keeps the display location of DA indicator **310** consistent with the physical location of external device **210**, e.g., moves DA indicator **310**'s location on display **306** leftwards relative to FIG. 3B.

(46) In some examples, while device **304** displays DA indicator **310** at the second location, device **304** samples a user speech input. For example, in FIG. 3B, user **302** issues a spoken request to the DA (e.g., "what's next on my calendar?") to initiate a DA interaction. The speech input does not include a spoken trigger for the DA, nor does user **302** provide any of the above-discussed explicit triggers to initiate the DA interaction.

(47) In some examples, while device **304** displays DA indicator **310** at the second location, device **304** determines whether a pose of user **302** corresponds to (e.g., whether user **302** faces) the second location. In some examples, determining that user **302**'s pose corresponds to the second location includes determining that user **302**'s gaze is directed at the second location, e.g., at the display of DA indicator **310**. For example, device **304** includes camera(s) configured to track user **302**'s eyes. Device **304** analyzes the captured camera data using eye tracking technologies to determine user **302**'s gaze direction. In some examples, determining that user **302**'s pose corresponds to the second

location includes detecting external device **210** using one or more sensors of device **304** when user **302** has the pose. For example, as discussed above, device **304** detects external device **210** within a particular region corresponding to the pose, e.g., within a front-facing region corresponding to the pose.

(48) In some examples, device **304** determines whether user **302**'s pose corresponds to the second location while sampling the user speech input. For example, device **304** determines whether user **302**'s pose corresponds to the second location while sampling the entirety of the speech input, while sampling a predetermined duration of the speech input, or while sampling a particular portion (e.g., start portion, middle portion, end portion) of the speech input. In some examples, in accordance with a determination that user **302**'s pose corresponds to the second location while sampling the user speech input, device **304** causes a response to the user speech input to be provided, e.g., by a DA.

(49) For example, because user **302** issues the spoken request “what's next on my calendar?” while facing the second location (e.g., while facing DA indicator **310**), device **304** causes a DA to respond to the request. In this manner, to initiate a DA interaction, user **302** can face a DA that appears to be present at a known location and issue a spoken request. User **302** is not required to provide an explicit trigger to initiate the interaction, as device **304** can determine user intent to interact via user pose and/or user gaze.

(50) In some examples, causing the response to be provided includes causing a DA implemented on external device **210** to provide a response using a speaker of external device **210**. For example, device **304** transmits the speech input “what's next on my calendar?” to external device **210** and instructs DA **212** to determine a response to the speech input. As shown in FIG. 3B, DA **212** determines the response “you have a meeting at 10 am” and causes external device **210** to audibly output the response.

(51) In some examples, causing the response to be provided includes providing, by a DA implemented on device **304**, the response using a speaker of device **304**. For example, DA **202** (implemented on device **304**) determines the response “you have a meeting at 10 am” and causes device **304**'s internal speaker(s) to output the response. In some examples, device **304** adjusts a characteristic of the response based on the second location and provides, using device **304**'s speaker(s), the adjusted response. For example, device **304** applies 3D-audio processing techniques on the response so the adjusted response appears to emanate from the location of DA indicator **310**.

(52) In some examples, device **304** ceases to display DA indicator **310** at the second location (including ceasing to display DA indicator **310** for any pose of user **302**). For example, device **304** ceases to display DA indicator **310** a predetermined duration after initially displaying DA indicator **310** at the second location. For example, device **304** ceases to display DA indicator **310** if device **304** does not sample speech input to initiate a DA interaction before the predetermined duration elapses. As another example, device **304** ceases to display DA indicator **310** responsive to receiving user input representing an instruction to dismiss the DA. Example user input includes speech input to dismiss the DA (e.g., “go away” or “I'm done”), button input to dismiss the DA, gesture input (e.g., predetermined gestures performed by user **302** to dismiss the DA, and motion input (e.g., predetermined motions detected by device **304** to dismiss the DA). In this manner, the DA can be dismissed, e.g., by entirely ceasing the display of DA indicator **310**, regardless of user **302**'s pose.

(53) In some examples, after ceasing to display DA indicator **310** at the second location (e.g., dismissing the DA), device **304** receives a user input. In some examples, in accordance with a determination that the user input satisfies the criterion for initiating the DA, device **304** displays (e.g., displays again) DA indicator **310** at the second location, e.g., if the second location is visible from user **302**'s current pose. For example, device **304** displays DA indicator **310** at the second location without first displaying DA indicator **310** at any other location. In this manner, if user **302** provides an explicit trigger to re-initiate the DA after DA dismissal, the display of DA indicator **310** remains anchored to the physical location of external device **210**, thereby enabling initiation of

future DA interactions without an explicit trigger.

(54) FIG. 4 is a flow diagram illustrating process 400 for digital assistant interaction, according to various examples. Process 400 is performed, for example, at a device (e.g., device 304) and using system 200. In process 400, some operations are, optionally, combined, the orders of some operations are, optionally, changed, and some operations are, optionally, omitted. In some examples, additional operations are performed in combination with process 400.

(55) In some examples, at block 402, while displaying, on a display of an electronic device (e.g., display 306 of device 304), an extended reality (XR) environment, a user input is received from a user.

(56) In some examples, at block 404, while displaying the XR environment, a user speech input is sampled with a microphone of the electronic device.

(57) In some examples, at block 406, it is determined whether the user input satisfies a criterion for initiating a digital assistant. In some examples, the user input satisfies the criterion for initiating the digital assistant if the user input includes a spoken trigger for the digital assistant.

(58) In some examples, at block 408, in accordance with a determination that the user input does not satisfy the criterion for initiating the digital assistant, initiating the digital assistant is foregone.

(59) In some examples, at block 410, in accordance with a determination that the user input satisfies the criterion for initiating the digital assistant, the digital assistant is initiated. In some examples, initiating the digital assistant includes displaying, within the XR environment, a digital assistant indicator (e.g., DA indicator 310) at a first location of the XR environment, as shown at block 412. In some examples, initiating the digital assistant includes, while displaying the digital assistant indicator at the first location, providing, by the digital assistant, a response to the user speech input, as shown at block 414.

(60) In some examples, at block 416, after providing the response, display of the digital assistant indicator at the first location is ceased. In some examples, ceasing to display the digital assistant indicator at the first location is performed automatically at a predetermined duration after providing the response. In some examples, after providing the response and while displaying the digital assistant indicator at the first location, a second user input representing an instruction to dismiss the display of the digital assistant indicator at the first location is received. In some examples, ceasing to display the digital assistant indicator at the first location is performed responsive to receiving the second user input. In some examples, the second user input includes gesture input or speech input.

(61) In some examples, at block 418, in accordance with ceasing to display the digital assistant indicator at the first location, the digital assistant indicator is displayed at a second location of the XR environment. The second location corresponds to a physical location of an external electronic device (e.g., external device 210) implementing a second digital assistant (e.g., DA 212). In some examples, the second location is above the physical location of the external electronic device and defined by the vertical axis of the external electronic device. In some examples, displaying the digital assistant indicator at the second location is performed in accordance with a determination that the second location is visible from a current pose of the user.

(62) In some examples, the display of the XR environment is updated based on an updated pose of the user. In some examples, displaying the digital assistant indicator at the second location includes maintaining the display of the digital assistant indicator at the second location while updating the display of the XR environment.

(63) In some examples, a physical representation of the external electronic device is displayed within the XR environment at the physical location. In some examples, a pose (e.g., head pose) of the user is determined (e.g., by device 304) and it is determined whether the external electronic device is visible from the determined pose. In some examples, displaying the physical representation of the external electronic device is performed in accordance with a determination that the external electronic device is visible from the determined pose. In some examples, determining that the external electronic device is visible from the determined pose includes

detecting the external electronic device using one or more sensors of the electronic device when the user has the determined pose.

(64) In some examples, while the digital assistant indicator is displayed at the second location, it is determined that a second pose of the user corresponds to the second location and a second user speech input is sampled with the microphone. In some examples, the second user speech input does not include a spoken trigger for the digital assistant. In some examples, determining that the second pose corresponds to the second location includes determining that a user gaze is directed at the second location. In some examples, determining that the second pose corresponds to the second location includes detecting the external electronic device using one or more sensors of the electronic device when the user has the second pose.

(65) In some examples, in accordance with a determination that the second pose corresponds to the second location while sampling the second user speech input, a second response to the second user speech input is caused to be provided.

(66) In some examples, causing the second response be provided includes causing the second digital assistant to provide the second response using a speaker of the external electronic device. In some examples, causing the second response be provided includes providing, by the digital assistant, the second response using a speaker of the electronic device. In some examples, providing, by the digital assistant, the second response includes: adjusting a characteristic of the second response based on the second location to obtain an adjusted response; and providing, using the speaker, the adjusted response.

(67) In some examples, display of the digital assistant indicator at the second location is ceased. In some examples, after ceasing to display the digital assistant indicator at the second location: a third user input is received; and in accordance with a determination that the third user input satisfies the criterion for initiating the digital assistant, the digital assistant indicator is displayed at the second location.

(68) The operations discussed above with respect to FIG. 4 are optionally implemented by the components depicted in FIG. 2, e.g., by system **100**, DA **202**, external device **210**, and DA **212**.

(69) In some examples, a computer-readable storage medium (e.g., a non-transitory computer readable storage medium) is provided, the computer-readable storage medium storing one or more programs for execution by one or more processors of an electronic device, the one or more programs including instructions for performing any of the methods or processes described herein.

(70) In some examples, an electronic device is provided that comprises means for performing any of the methods or processes described herein.

(71) In some examples, an electronic device is provided that comprises a processing unit configured to perform any of the methods or processes described herein.

(72) In some examples, an electronic device is provided that comprises one or more processors and memory storing one or more programs for execution by the one or more processors, the one or more programs including instructions for performing any of the methods or processes described herein.

(73) Various processes described herein contemplate the option of obtaining and using a user's personal information. For example, such personal information can be used to provide efficient DA interactions. However, if such personal information is obtained, such information should be obtained with the user's informed consent. Specifically, the user should have knowledge of and control over how devices use their personal information.

(74) Appropriate parties will use personal information only for reasonable and legitimate purposes. These parties will abide by privacy policies and practices that at least comply with appropriate laws and regulations. Further, such policies should be user-accessible, well-established, and recognized as compliant with (or to exceed) governmental/industry standards. Additionally, these parties will not sell, distribute, or otherwise share personal information for any unreasonable or illegitimate purposes.

(75) Users may also limit the degree to which such parties may access or otherwise obtain personal information. For instance, user settings or other preferences can be changed to allow users to decide whether their personal information is accessible by various entities. Furthermore, while some features described herein are described to include use of personal information, various aspects of these features may be implemented without needing use of such information. For example, if user location history is collected, this information can be obscured or otherwise generalized such that the information does not identify the corresponding user.

Claims

1. A non-transitory computer-readable storage medium storing one or more programs, the one or more programs comprising instructions, which when executed by one or more processors of an electronic device having a display and a microphone, cause the electronic device to: while displaying, on the display, an extended reality (XR) environment: receive, from a user, a user input; sample, with the microphone, a user speech input; in accordance with a determination that the user input satisfies a criterion for initiating a digital assistant, initiate the digital assistant, including: displaying, within the XR environment, a digital assistant indicator at a first location of the XR environment; and while displaying the digital assistant indicator at the first location, provide, by the digital assistant, a response to the user speech input; after providing the response, cease to display the digital assistant indicator at the first location; in accordance with ceasing to display the digital assistant indicator at the first location, display the digital assistant indicator at a second location of the XR environment, the second location corresponding to a physical location of an external electronic device implementing a second digital assistant; while displaying the digital assistant indicator at the second location: determine that a pose of the user corresponds to the second location; and sample, with the microphone, a second user speech input; and in accordance with a determination that the pose corresponds to the second location while sampling the second user speech input, cause a second response to the second user speech input to be provided.
2. The non-transitory computer-readable storage medium of claim 1, wherein the user input satisfies the criterion for initiating the digital assistant if the user input includes a spoken trigger for the digital assistant.
3. The non-transitory computer-readable storage medium of claim 1, wherein ceasing to display the digital assistant indicator at the first location is performed automatically at a predetermined duration after providing the response.
4. The non-transitory computer-readable storage medium of claim 1, wherein the one or more programs further include instructions, which when executed by the one or more processors, cause the electronic device to: after providing the response and while displaying the digital assistant indicator at the first location, receive a second user input representing an instruction to dismiss the display of the digital assistant indicator at the first location, wherein ceasing to display the digital assistant indicator at the first location is performed responsive to receiving the second user input.
5. The non-transitory computer-readable storage medium of claim 4, wherein the second user input includes gesture input or speech input.
6. The non-transitory computer-readable storage medium of claim 1, wherein displaying the digital assistant indicator at the second location is performed in accordance with a determination that the second location is visible from a current pose of the user.
7. The non-transitory computer-readable storage medium of claim 1, wherein the one or more programs further include instructions, which when executed by the one or more processors, cause the electronic device to: display, within the XR environment, a physical representation of the external electronic device at the physical location.
8. The non-transitory computer-readable storage medium of claim 7, wherein the one or more programs further include instructions, which when executed by the one or more processors, cause

the electronic device to: determine a second pose of the user; and determine whether the external electronic device is visible from the determined second pose, wherein displaying the physical representation of the external electronic device is performed in accordance with a determination that the external electronic device is visible from the determined second pose.

9. The non-transitory computer-readable storage medium of claim 8, wherein determining that the external electronic device is visible from the determined second pose includes detecting the external electronic device using one or more sensors of the electronic device when the user has the determined second pose.

10. The non-transitory computer-readable storage medium of claim 1, wherein the second location is above the physical location of the external electronic device and defined by the vertical axis of the external electronic device.

11. The non-transitory computer-readable storage medium of claim 1, wherein causing the second response to be provided includes causing the second digital assistant to provide the second response using a speaker of the external electronic device.

12. The non-transitory computer-readable storage medium of claim 1, wherein causing the second response to be provided includes providing, by the digital assistant, the second response using a speaker of the electronic device.

13. The non-transitory computer-readable storage medium of claim 12, wherein providing, by the digital assistant, the second response includes: adjusting a characteristic of the second response based on the second location to obtain an adjusted response; and providing, using the speaker, the adjusted response.

14. The non-transitory computer-readable storage medium of claim 1, wherein the second user speech input does not include a spoken trigger for the digital assistant.

15. The non-transitory computer-readable storage medium of claim 1, wherein determining that the pose corresponds to the second location includes determining that a user gaze is directed at the second location.

16. The non-transitory computer-readable storage medium of claim 1, wherein determining that the pose corresponds to the second location includes detecting the external electronic device using one or more sensors of the electronic device when the user has the pose.

17. The non-transitory computer-readable storage medium of claim 1, wherein the one or more programs further include instructions, which when executed by the one or more processors, cause the electronic device to: cease to display the digital assistant indicator at the second location; and after ceasing to display the digital assistant indicator at the second location: receive a third user input; and in accordance with a determination that the third user input satisfies the criterion for initiating the digital assistant, display the digital assistant indicator at the second location.

18. The non-transitory computer-readable storage medium of claim 1, wherein the one or more programs further include instructions, which when executed by the one or more processors, cause the electronic device to: update the display of the XR environment based on an updated pose of the user; and wherein displaying the digital assistant indicator at the second location includes maintaining the display of the digital assistant indicator at the second location while updating the display of the XR environment.

19. An electronic device comprising: a display; a microphone; one or more processors; a memory; and one or more programs, wherein the one or more programs are stored in the memory and configured to be executed by the one or more processors, the one or more programs including instructions for: while displaying, on the display, an extended reality (XR) environment: receiving, from a user, a user input; sampling, with the microphone, a user speech input; in accordance with a determination that the user input satisfies a criterion for initiating a digital assistant, initiating the digital assistant, including: displaying, within the XR environment, a digital assistant indicator at a first location of the XR environment; and while displaying the digital assistant indicator at the first location, providing, by the digital assistant, a response to the user speech input; after providing the

response, ceasing to display the digital assistant indicator at the first location; in accordance with ceasing to display the digital assistant indicator at the first location, displaying the digital assistant indicator at a second location of the XR environment, the second location corresponding to a physical location of an external electronic device implementing a second digital assistant; while displaying the digital assistant indicator at the second location: determining that a pose of the user corresponds to the second location; and sampling, with the microphone, a second user speech input; and in accordance with a determination that the pose corresponds to the second location while sampling the second user speech input, causing a second response to the second user speech input to be provided.

20. The electronic device of claim 19, wherein the user input satisfies the criterion for initiating the digital assistant if the user input includes a spoken trigger for the digital assistant.

21. The electronic device of claim 19, wherein ceasing to display the digital assistant indicator at the first location is performed automatically at a predetermined duration after providing the response.

22. The electronic device of claim 19, wherein the one or more programs further include instructions for: after providing the response and while displaying the digital assistant indicator at the first location, receiving a second user input representing an instruction to dismiss the display of the digital assistant indicator at the first location, wherein ceasing to display the digital assistant indicator at the first location is performed responsive to receiving the second user input.

23. The electronic device of claim 19, wherein the one or more programs further include instructions for: displaying, within the XR environment, a physical representation of the external electronic device at the physical location.

24. The electronic device of claim 19, wherein causing the second response to be provided includes causing the second digital assistant to provide the second response using a speaker of the external electronic device.

25. The electronic device of claim 19, wherein causing the second response to be provided includes providing, by the digital assistant, the second response using a speaker of the electronic device.

26. The electronic device of claim 25, wherein providing, by the digital assistant, the second response includes: adjusting a characteristic of the second response based on the second location to obtain an adjusted response; and providing, using the speaker, the adjusted response.

27. The electronic device of claim 19, wherein the second user speech input does not include a spoken trigger for the digital assistant.

28. The electronic device of claim 19, wherein determining that the pose corresponds to the second location includes determining that a user gaze is directed at the second location.

29. The electronic device of claim 19, wherein determining that the pose corresponds to the second location includes detecting the external electronic device using one or more sensors of the electronic device when the user has the pose.

30. The electronic device of claim 19, wherein the one or more programs further include instructions for: ceasing to display the digital assistant indicator at the second location; and after ceasing to display the digital assistant indicator at the second location: receiving a third user input; and in accordance with a determination that the third user input satisfies the criterion for initiating the digital assistant, displaying the digital assistant indicator at the second location.

31. The electronic device of claim 19, wherein the one or more programs further include instructions for: updating the display of the XR environment based on an updated pose of the user; and wherein displaying the digital assistant indicator at the second location includes maintaining the display of the digital assistant indicator at the second location while updating the display of the XR environment.

32. A method, comprising: at an electronic device having one or more processors, memory, a microphone, and a display: while displaying, on the display, an extended reality (XR) environment: receiving, from a user, a user input; sampling, with the microphone, a user speech input; in

accordance with a determination that the user input satisfies a criterion for initiating a digital assistant, initiating the digital assistant, including: displaying, within the XR environment, a digital assistant indicator at a first location of the XR environment; and while displaying the digital assistant indicator at the first location, providing, by the digital assistant, a response to the user speech input; after providing the response, ceasing to display the digital assistant indicator at the first location; in accordance with ceasing to display the digital assistant indicator at the first location, displaying the digital assistant indicator at a second location of the XR environment, the second location corresponding to a physical location of an external electronic device implementing a second digital assistant; while displaying the digital assistant indicator at the second location: determining that a pose of the user corresponds to the second location; and sampling, with the microphone, a second user speech input; and in accordance with a determination that the pose corresponds to the second location while sampling the second user speech input, causing a second response to the second user speech input to be provided.

33. The method of claim 32, wherein the user input satisfies the criterion for initiating the digital assistant if the user input includes a spoken trigger for the digital assistant.

34. The method of claim 32, wherein ceasing to display the digital assistant indicator at the first location is performed automatically at a predetermined duration after providing the response.

35. The method of claim 32, further comprising: after providing the response and while displaying the digital assistant indicator at the first location, receiving a second user input representing an instruction to dismiss the display of the digital assistant indicator at the first location, wherein ceasing to display the digital assistant indicator at the first location is performed responsive to receiving the second user input.

36. The method of claim 32, further comprising: displaying, within the XR environment, a physical representation of the external electronic device at the physical location.

37. The method of claim 32, wherein causing the second response to be provided includes causing the second digital assistant to provide the second response using a speaker of the external electronic device.

38. The method of claim 32, wherein causing the second response to be provided includes providing, by the digital assistant, the second response using a speaker of the electronic device.

39. The method of claim 38, wherein providing, by the digital assistant, the second response includes: adjusting a characteristic of the second response based on the second location to obtain an adjusted response; and providing, using the speaker, the adjusted response.

40. The method of claim 32, wherein the second user speech input does not include a spoken trigger for the digital assistant.

41. The method of claim 32, wherein determining that the pose corresponds to the second location includes determining that a user gaze is directed at the second location.

42. The method of claim 32, wherein determining that the pose corresponds to the second location includes detecting the external electronic device using one or more sensors of the electronic device when the user has the pose.

43. The method of claim 32, further comprising: ceasing to display the digital assistant indicator at the second location; and after ceasing to display the digital assistant indicator at the second location: receiving a third user input; and in accordance with a determination that the third user input satisfies the criterion for initiating the digital assistant, displaying the digital assistant indicator at the second location.

44. The method of claim 32, further comprising: updating the display of the XR environment based on an updated pose of the user; and wherein displaying the digital assistant indicator at the second location includes maintaining the display of the digital assistant indicator at the second location while updating the display of the XR environment.
